THE STRENGTHENED SZEGED–WIENER CONJECTURE

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Abstract. Bonamy, Knor, Lužar, Pinlou and Škrekovski conjectured in 2017 that every 2-connected graph G of order $n \geq 10$, other than K_n, K_n^2 and K_n^{n-2}, satisfies

$$\mathrm{Sz}(G) - W(G) \geq 2n.$$

We prove the conjecture. The argument completes the dominated-vertex induction introduced in their work. Complete deletions are handled by an exact formula for K_n^t; the two exceptional 2-connected deletions are treated by direct true-twin counts; and non-2-connected deletions are reduced to a short list of small block-pieces. The finite part of the proof is confined to graphs of order at most ten and is verified by an exact, reproducible certificate supplied with the paper. No floating-point computation or graph-isomorphism reduction is used in the critical order-ten check.

1. Introduction

For a connected graph G, the Wiener index is

$$W(G) = \sum_{\{a,b\} \subseteq V(G)} d_G(a, b),$$

where d_G is graph distance. If $e = uv$ is an edge, let $n_e(u)$ be the number of vertices strictly closer to u than to v, and define $n_e(v)$ analogously. The Szeged index is

$$\mathrm{Sz}(G) = \sum_{uv \in E(G)} n_{uv}(u) n_{uv}(v).$$

We write

$$\eta(G) = \mathrm{Sz}(G) - W(G).$$

Bonamy, Knor, Lužar, Pinlou and Škrekovski proved that every 2-connected non-complete graph of order n satisfies $\eta(G) \geq 2n - 6$ [1, Theorem 2]. Their stronger Theorem 4, together with Lemma 11, identifies the equality graphs as K_n^2 and K_n^{n-2} and gives the bound $2n - 5$ after excluding them [1, Theorem 4 and Lemma 11]. They then proposed the following strengthening [1, Conjecture 5].

For $1 \leq t \leq n - 1$, let K_n^t denote the graph obtained from K_{n-1} by adding one vertex adjacent to exactly t vertices of the old clique. Thus $K_n^{n-1} = K_n$.

Theorem 1.1. *Let G be a 2-connected graph of order $n \geq 10$. If*

$$G \ncong K_n, \qquad G \ncong K_n^2, \qquad G \ncong K_n^{n-2},$$

then

$$\eta(G) \geq 2n.$$

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The proof stays close to the mechanism of [1]. The useful point is that a putative counterexample must contain a vertex whose closed neighbourhood is contained in another closed neighbourhood. Deleting such a vertex either preserves 2-connectivity, or forces a rigid block decomposition. The former case has two exceptional subcases not covered by a direct induction; the latter case requires stronger low-order estimates than those used in [1, Theorem 4]. These are the two places where new input is needed.

The finite verification appears only after the structural reductions have been proved. We state the exact finite universes in Section 5; the supplementary certificate merely evaluates the definitions on those universes.

2. Good edges and dominated vertices

Throughout, a *dominated pair* (u, v) consists of two *distinct* vertices $u \neq v$ such that $N[u] \subseteq N[v]$. Every occurrence of a dominated pair or of the condition $N[u] \subseteq N[v]$ in a deletion lemma, proposition, or its proof is subject to this distinctness convention. In particular, the vertex deleted in $G - u$ is different from its dominator v.

We use the good-edge formulation of [1]. For an unordered pair $\{a, b\}$, an edge xy is *good* for $\{a, b\}$ if, after possibly interchanging x and y,

$$d_G(a, x) < d_G(a, y), \qquad d_G(b, y) < d_G(b, x).$$

Let $g_G(a, b)$ be the number of good edges for $\{a, b\}$ and set

$$\pi_G(a, b) = g_G(a, b) - d_G(a, b).$$

Proposition 2.1 (Bonamy–Knor–Lužar–Pinlou–Škrekovski)**.** *For every connected graph G,*

$$\eta(G) = \sum_{\{a,b\} \subseteq V(G)} \pi_G(a, b),$$

and every summand is non-negative.

Proof. This is [1, Proposition 6]. □

We shall also use the following three published lemmas. They are quoted with all hypotheses because the induction below depends on them exactly as stated.

Lemma 2.2 (Dominated deletion)**.** *Let G be 2-connected. Suppose that $u, v \in V(G)$ satisfy $N[u] \subseteq N[v]$, and that $G - u$ is 2-connected and non-complete. Then*

$$\eta(G) - \eta(G - u) \geq 2.$$

Proof. This is [1, Lemma 7]. □

Lemma 2.3 (Structure after a non-2-connected deletion)**.** *Let G be a 2-connected non-complete graph distinct from K_4^2. Assume that at least one pair u, v satisfies $N[u] \subseteq N[v]$, and that for every such pair the graph $G - u$ is complete or is not 2-connected. Then, for every such pair, $G - u$ is not 2-connected. Moreover, the dominator v is the unique cut vertex of $G - u$, and no block of $G - u$ is isomorphic to K_t, K_t^2, or K_t^{t-2}.*

Proof. The first and last assertions are [1, Lemma 9]; the proof there also shows explicitly that v is the unique cut vertex of $G - u$. □

Lemma 2.4 (No dominated pair)**.** *Let G be a 2-connected non-complete graph distinct from C_5. If no two vertices u, v satisfy $N[u] \subseteq N[v]$, then*

$$\eta(G) \geq 2|V(G)|.$$

Proof. This is the final conclusion of [1, Lemma 10]. □

Two elementary observations will make the block argument transparent.

Lemma 2.5. *If $N[u] \subseteq N[v]$, then $uv \in E(G)$ and*

$$\pi_G(u, v) = 0.$$

Proof. The inclusion contains $u \in N[v]$, so uv is an edge. For $z = v$ the inequality $d(v, z) \leq d(u, z)$ is immediate. If $z \notin \{u, v\}$, a shortest u–z path begins with a neighbour of u; that neighbour also lies in $N(v)$, and hence again $d(v, z) \leq d(u, z)$. Since uv is an edge, $d(u, z) \leq d(v, z) + 1$.

The edge uv is good for $\{u, v\}$. Suppose another edge xy were good, oriented so that $d(u, x) < d(u, y)$. If $x = u$, then the strict inequality at v forces $y = v$, so the edge is uv. Hence for another good edge we have $x \neq u$, and the inequality above gives $d(u, x) \geq d(v, x)$. Adjacency gives $d(u, y) = d(u, x) + 1$, while the edge uv gives

$$d(v, y) \geq d(u, y) - 1 = d(u, x) \geq d(v, x),$$

contradicting the reverse strict inequality required at v. Thus $g(u, v) = 1 = d(u, v)$. □

Lemma 2.6 (Encoding a dominated extension)**.** *Let $H = G - u$ and suppose $N_G[u] \subseteq N_G[v]$. Then*

$$N_G(u) = \{v\} \cup T$$

for some $T \subseteq N_H(v)$. If G is 2-connected, then $T \neq \varnothing$. Conversely, adjoining to any graph H a new vertex u with neighbourhood $\{v\} \cup T$, where $T \subseteq N_H(v)$, produces $N[u] \subseteq N[v]$. If, in addition, H is 2-connected and $T \neq \varnothing$, then the resulting graph is 2-connected.

Proof. The closed-neighbourhood inclusion gives $uv \in E(G)$, and every other neighbour of u must be adjacent to v. In a 2-connected graph $\deg(u) \geq 2$, so T is non-empty. The converse inclusion is immediate. For the final assertion, deleting u leaves H; deleting a vertex of H leaves the connected graph $H - x$, and at least one of the two distinct neighbours v and a vertex of T remains adjacent to u. Thus deletion of any single vertex leaves a connected graph. □

3. The exceptional 2-connected deletions

We first remove the case in which deleting the dominated vertex leaves a complete graph.

Proposition 3.1. *For $2 \leq t \leq n - 2$,*

$$\eta(K_n^t) = 2(t - 1)(n - t - 1).$$

Proof. Let x be the added vertex, let $A = N(x)$ with $|A| = t$, and let B be the remaining $s = n - 1 - t$ vertices of the old clique. Then

$$W(K_n^t) = \binom{n-1}{2} + t + 2s.$$

Every edge inside the old clique contributes 1 to the Szeged index, except that each of the ts edges between A and B receives one additional unit from x. Each of the t edges from x to A contributes $s + 1$. Hence

$$\mathrm{Sz}(K_n^t) = \binom{n-1}{2} + ts + t(s + 1),$$

and subtraction gives the formula. □

Corollary 3.2. *Let $n \geq 10$. If G is 2-connected, $N[u] \subseteq N[v]$, and $G - u$ is complete, then either $G \cong K_n, K_n^2, K_n^{n-2}$ or $\eta(G) \geq 2n$.*

Proof. We have $G \cong K_n^t$. The cases $t = 1$ and $t = n-1$ are, respectively, not 2-connected and complete. After excluding $t = 2, n-2$, it remains to consider $3 \leq t \leq n-3$. The expression in Proposition 3.1 is concave in t, so its minimum on this interval occurs at an endpoint and equals $4n - 16 \geq 2n$. □

The other obstruction to the direct induction is that $G - u$ may itself be one of the two equality graphs identified by [1, Theorem 4 and Lemma 11]. Their symmetry makes an exact calculation short. We record the counting device once.

Lemma 3.3 (True-twin quotient count)**.** *Suppose $V(G)$ is partitioned into non-empty classes $V_1, \ldots, V_r$ such that each V_i is a clique of true twins (singletons are allowed), and between two distinct classes all edges or no edges are present. Let $m_i = |V_i|$, let Q be the quotient graph, and write D_{ij} for distance in Q. Then*

$$W(G) = \sum_i \binom{m_i}{2} + \sum_{i<j} m_i m_j D_{ij}.$$

For an edge ij of Q, put

$$A_{ij} = 1 + \sum_{h \notin \{i,j\} \,:\, D_{hi} < D_{hj}} m_h, \qquad B_{ij} = 1 + \sum_{h \notin \{i,j\} \,:\, D_{hj} < D_{hi}} m_h.$$

Then

$$\mathrm{Sz}(G) = \sum_i \binom{m_i}{2} + \sum_{ij \in E(Q)} m_i m_j A_{ij} B_{ij}.$$

Proof. Distances are constant between two distinct classes and equal to the quotient distance. An internal edge of a true-twin clique has one vertex on each strict side and all other vertices are equidistant, so its Szeged contribution is 1. For a cross edge between V_i and V_j, the other vertices in $V_i \cup V_j$ are equidistant from its endpoints, while every third class lies entirely on one strict side or is equidistant according to the quotient distances. This gives the displayed formulas. □

3.1. **Deletion to K_{n-1}^2.** Assume $H = G - u \cong K_{n-1}^2$. Write the old clique as

$$Q = \{a, b\} \cup R, \qquad |R| = n - 4,$$

and let x be the exceptional vertex, adjacent precisely to a, b in H. Set

$$\varepsilon = \mathbf{1}_{ux \in E(G)}, \qquad k = |N(u) \cap \{a, b\}|, \qquad \ell = |N(u) \cap R|.$$

Lemma 3.4. *If G is 2-connected and $N[u] \subseteq N[v]$ for some $v \in V(H)$, then (ε, k) is one of*

$$(0,0), (0,1), (0,2), (1,1), (1,2),$$

with the ranges for ℓ shown in Table 1.

Proof. If $v = x$, then u is adjacent to x, to no vertex of R, and to at least one of a, b. If $v \in \{a, b\}$, that vertex is dominating in H, so the only condition is that it be adjacent to u, hence $k \geq 1$. If $v \in R$, then $x \notin N[v]$, so $\varepsilon = 0$. These three possibilities give exactly the five rows. The lower bounds on ℓ are precisely $\deg(u) \geq 2$; the upper bound is $|R| = n - 4$. □

For an interior value $0 < \ell < n-4$, split $\{a, b\}$ into the selected and unselected classes S, T of sizes $k, 2-k$, and split R into the selected and unselected classes L, M of sizes $\ell, n-4-\ell$. The classes S, T, L, M form a clique; x is adjacent exactly to $S \cup T$; and u is adjacent to $S \cup L$ and, when $\varepsilon = 1$, to x. Applying Lemma 3.3 gives the following exact expressions. When L or M is empty, the same endpoint values are obtained by omitting that class and repeating the quotient count; thus no polynomial extrapolation is being used.

TABLE 1. The case $G - u \cong K^2_{n-1}$. The last column excludes the endpoint that is isomorphic to K^2_n.

(ε, k)	admissible ℓ	$\eta(G)$	non-exceptional minimum
$(0,0)$	$2 \le \ell \le n-4$	$-2\ell^2 + (2n+1)\ell - 5$	$4n-11$
$(0,1)$	$1 \le \ell \le n-4$	$-2\ell^2 + (2n-5)\ell + 2n - 8$	$4n-15$
$(0,2)$	$0 \le \ell \le n-4$	$-2\ell^2 + (2n-10)\ell + 4n - 14$	$4n-14$
$(1,1)$	$0 \le \ell \le n-4$	$-3\ell^2 + (3n-5)\ell + 3n - 10$	$3n-10$
$(1,2)$	$0 \le \ell \le n-4$	$-3\ell^2 + (3n-12)\ell + 4n - 16$	$4n-16$

Each row is a concave quadratic in ℓ. The only exceptional endpoint is $(\varepsilon, k, \ell) = (0, 2, n-4)$, which is K^2_n and has value $2n-6$. Direct endpoint counts give, when $\ell = n-4$, respectively

$$9n-41, \quad 5n-20, \quad 2n-6, \quad 10n-38, \quad 4n-16,$$

and when $\ell = 0$ is admissible they give

$$4n-14, \quad 3n-10, \quad 4n-16$$

for the last three relevant rows. Hence the minima in Table 1 follow.

For illustration, in the extremal row $(\varepsilon, k) = (1, 1)$ and $0 < \ell < n-4$, the quotient classes have sizes

$$1, 1, 1, 1, \ell, n-4-\ell$$

for $\{u\}, \{x\}, S, T, L, M$. Substitution in Lemma 3.3 gives

$$W(G) = -\ell^2 + n\ell - 5\ell + 6n - 17 + \binom{\ell}{2} + \binom{n-4-\ell}{2},$$
$$\mathrm{Sz}(G) = -4\ell^2 + 4n\ell - 10\ell + 9n - 27 + \binom{\ell}{2} + \binom{n-4-\ell}{2},$$

so their difference is the fourth row of Table 1. The other rows are obtained by the same quotient calculation; the tables record the full arithmetic needed later.

Proposition 3.5. *Let $n \ge 10$. Suppose G is 2-connected, $N[u] \subseteq N[v]$, and $G - u \cong K^2_{n-1}$. If $G \ncong K^2_n$, then*

$$\eta(G) \ge 3n - 10 \ge 2n.$$

3.2. **Deletion to K^{n-3}_{n-1}.** Now let $H = G - u \cong K^{n-3}_{n-1}$. Write the old clique as $Q = A \cup \{z\}$ with $|A| = n-3$, and let x be adjacent to every vertex of A and not to z. Set

$$\varepsilon = \mathbf{1}_{ux \in E(G)}, \qquad \delta = \mathbf{1}_{uz \in E(G)}, \qquad \ell = |N(u) \cap A|.$$

If the dominator lies in A, it must be one of the ℓ selected vertices and imposes no further restriction. If it is x, then $(\varepsilon, \delta) = (1, 0)$; if it is z, then $(\varepsilon, \delta) = (0, 1)$. Consequently all four pairs $(0,0), (0,1), (1,0), (1,1)$ occur, with $\ell \ge 2$ in the first row and $\ell \ge 1$ in the others.

Splitting A into selected and unselected true-twin classes and applying Lemma 3.3 gives Table 2. The endpoint $\ell = n-3$ is checked directly after deleting the empty unselected class.

TABLE 2. The case $G-u \cong K_{n-1}^{n-3}$. The exceptional endpoint in the last row is K_n^{n-2}.

(ε,δ)	admissible ℓ	$\eta(G)$	non-exceptional minimum
$(0,0)$	$2 \le \ell \le n-3$	$-2\ell^2+2n\ell-6$	$4n-14$
$(0,1)$	$1 \le \ell \le n-3$	$-2\ell^2+(2n-5)\ell+3n-13$	$4n-16$
$(1,0)$	$1 \le \ell \le n-3$	$-2\ell^2+(2n-5)\ell+3n-13$	$4n-16$
$(1,1)$	$1 \le \ell \le n-3$	$-2\ell^2+(2n-12)\ell+8n-24$	$4n-8$

At $\ell = n-3$ the four rows have values

$$6n-24, \quad 4n-16, \quad 4n-16, \quad 2n-6,$$

respectively; the last graph is exactly K_n^{n-2}. Concavity again gives the final column of Table 2.

Proposition 3.6. *Let $n \ge 10$. Suppose G is 2-connected, $N[u] \subseteq N[v]$, and $G-u \cong K_{n-1}^{n-3}$. If $G \ncong K_n^{n-2}$, then*

$$\eta(G) \ge 4n-16 \ge 2n.$$

4. Block pieces

Suppose now that G is 2-connected, $N[u] \subseteq N[v]$, and $G-u$ is not 2-connected. Under the hypotheses of Lemma 2.3, let $C_1, \dots, C_k$ be the blocks of $G-u$. The vertex v is their unique common cut vertex. Put

$$G_i = G[V(C_i) \cup \{u\}], \qquad s_i = |V(G_i)|.$$

Since G is 2-connected, u has in every C_i a neighbour $w_i \ne v$; otherwise v would be a cut vertex of G. Hence each G_i is 2-connected by Lemma 2.6, applied to the block C_i. By Lemma 2.3, each C_i is non-complete and is not K_t^2 or K_t^{t-2}; in particular $|C_i| \ge 4$ and $s_i \ge 5$.

The proof of [1, Theorem 4] observes in precisely this situation that every G_i is a proper 2-connected subgraph and is neither complete nor isomorphic to $K_{s_i}^2$ or $K_{s_i}^{s_i-2}$. For completeness, we isolate the short reason.

Lemma 4.1. *Each G_i is 2-connected and*

$$G_i \ncong K_{s_i}, \qquad G_i \ncong K_{s_i}^2, \qquad G_i \ncong K_{s_i}^{s_i-2}.$$

Proof. The 2-connectivity was noted above. If G_i were complete, then C_i would be complete. In $K_{s_i}^2$, deleting the degree-2 vertex leaves K_{s_i-1}; deleting one of the s_i-3 old non-universal clique vertices leaves $K_{s_i-1}^2$; and deleting either universal vertex makes the degree-2 vertex pendant, so the deletion is not 2-connected. Hence a deletion that remains 2-connected is complete or $K_{s_i-1}^2$.

In $K_{s_i}^{s_i-2}$, deleting either of the two non-universal vertices leaves K_{s_i-1}, whereas deleting a universal vertex leaves $K_{s_i-1}^{s_i-3}$. Thus, again, every 2-connected deletion is complete or one of the two families forbidden for C_i by Lemma 2.3. This is the same case analysis used in [1, proof of Theorem 4]. □

We shall use the following small-piece lower bounds, proved in Section 5 by an exact finite certificate:

s	5	6	7	8	9
$L(s)$	10	8	11	14	17.

(1)

Here $L(s)$ is a lower bound for every s-vertex 2-connected graph of the form in Lemma 2.6 whose deletion graph is a non-special block. For $s \geq 10$ occurring inside a minimal counterexample, the inductive bound will be $L(s) = 2s$.

Lemma 4.2 (Block assembly)**.** *With the notation above,*

$$\eta(G) \geq \sum_{i=1}^{k} \eta(G_i) + k(k-1).$$

Proof. The pieces meet only in u and v. A path between two vertices of a fixed G_i that leaves G_i can enter another block only through u or v; since uv is already an edge, replacing the outside detour by uv never increases the length. Hence distances inside G_i are the same in G. Every edge of G_i that is good for a pair inside G_i therefore remains good in G. The only pair counted in more than one G_i is $\{u, v\}$, and its contribution is zero by Lemma 2.5. Thus the contributions from within the pieces sum to at least $\sum_i \eta(G_i)$.

For $i \neq j$, the vertices w_i, w_j lie in distinct blocks of $G - u$ and so are non-adjacent. Both are adjacent to u and, because $N[u] \subseteq N[v]$, also to v. The four edges w_iu, uw_j, w_iv, vw_j are good for $\{w_i, w_j\}$, while $d(w_i, w_j) = 2$. Hence

$$\pi_G(w_i, w_j) \geq 2.$$

Summing over the $\binom{k}{2}$ unordered pairs of blocks gives the extra term $2\binom{k}{2} = k(k-1)$. □

Because u and v occur in every piece,

$$\sum_{i=1}^{k} s_i = n + 2(k-1).$$

Define $d(s) = L(s) - 2s$. From (1),

$$d(5) = 0, \quad d(6) = -4, \quad d(7) = -3, \quad d(8) = -2, \quad d(9) = -1.$$

For a piece of order at least ten inside a minimal counterexample we shall have $d(s) \geq 0$. Therefore Lemma 4.2 yields

$$\eta(G) \geq 2n + (k-1)(k+4) + \sum_{i=1}^{k} d(s_i). \tag{2}$$

Lemma 4.3. *If $k \geq 3$, then the right-hand side of (2) is at least $2n+2$. If $k = 2$ and $n \geq 12$, it is at least $2n$. For $n = 10, 11$ the only combinations not settled by (2) are*

$$n = 10 : \ (s_1, s_2) = (6, 6), \qquad n = 11 : \ (s_1, s_2) = (6, 7).$$

Proof. Every deficit is at least -4. For $k \geq 3$, (2) is at least

$$2n + (k-1)(k+4) - 4k = 2n + k^2 - k - 4 \geq 2n + 2.$$

If $k = 2$ and $n \geq 12$, then $s_1 + s_2 = n + 2 \geq 14$ with each $s_i \geq 5$. Inspection of the five displayed deficits, together with $d(s) \geq 0$ for $s \geq 10$, gives $d(s_1) + d(s_2) \geq -6$, which cancels the term $(k-1)(k+4) = 6$.

For $n = 10$, $s_1 + s_2 = 12$; the pair $(5, 7)$ is already safe and only $(6, 6)$ remains. For $n = 11$, $s_1 + s_2 = 13$; $(5, 8)$ is safe and only $(6, 7)$ remains. □

5. The finite certificate

This section states exactly what is verified computationally and why those finite universes cover every case left by the preceding lemmas. All values of W, Sz and η are computed as integers from their definitions.

The input small graphs are taken from the Atlas of Graphs of Read and Wilson [2]. The supplied snapshot was extracted with `graph_atlas_g` from NetworkX 3.6.1 [3]; the NetworkX documentation identifies this routine as the list of all graphs on at most seven vertices named in the Atlas. The certificate contains the fixed snapshot and its SHA256 digest. Regeneration is optional: the C++ verifier runs directly from the snapshot.

For a graph C, a vertex $v \in V(C)$ and a non-empty set $T \subseteq N_C(v)$, write

$$P(C, v, T)$$

for the graph obtained by adjoining a new vertex u with $N(u) = \{v\} \cup T$. By Lemma 2.6, every block-piece in the proof has this form.

Proposition 5.1 (Small-piece certificate)**.** *Let $5 \le s \le 9$. Let C be a 2-connected graph of order $s-1$ that is not isomorphic to K_{s-1}, K^2_{s-1}, or K^{s-3}_{s-1}. For every $v \in V(C)$ and every non-empty $T \subseteq N_C(v)$,*

$$\eta(P(C, v, T)) \ge L(s),$$

where the exact minima are

$$L(5) = 10, \quad L(6) = 8, \quad L(7) = 11, \quad L(8) = 14, \quad L(9) = 17.$$

Proof. For $5 \le s \le 8$, the deletion block C has $4 \le |C| \le 7$. The certificate runs over every 2-connected Atlas representative C of that order other than K_t, K^2_t, K^{t-2}_t, every choice of v, and every non-empty $T \subseteq N_C(v)$. The numbers of generated parameter choices and the resulting minima are

s	5	6	7	8
$\#P(C, v, T)$	12	209	3258	48865
$\min \eta$	10	8	11	14.

For $s = 9$, let C have order eight. If $\eta(C) \ge 15$, then Lemma 2.2 gives $\eta(P(C, v, T)) \ge 17$. It remains only to consider 2-connected, non-special C with $\eta(C) \le 14$. Every 2-connected eight-vertex graph becomes connected after deletion of any vertex, so it is obtained by adding one non-isolated vertex to a connected seven-vertex Atlas representative. The certificate generates all such extensions; 65 generated occurrences survive the stated filters. Every dominated extension of these occurrences is then checked, and the minimum is 17. Repeated isomorphic occurrences are deliberately retained. □

We next isolate the two small gluings left by Lemma 4.3.

Proposition 5.2 (Glue certificate)**.** *In the block-decomposition setting of Section 4, suppose $k = 2$. If $n = 10$ and $(s_1, s_2) = (6, 6)$, then $\eta(G) \ge 20$; if $n = 11$ and $(s_1, s_2) = (6, 7)$, then $\eta(G) \ge 23$.*

Proof. In the first case Lemma 4.2 already gives $\eta(G) \ge 20$ whenever $\eta(G_1) + \eta(G_2) \ge 18$. Since $L(6) = 8$, only 6-vertex pieces of values 8 or 9 need be retained, and only pairs with sum below 18 need be glued. The piece generation in Proposition 5.1 yields 15 such piece occurrences; there are 105 relevant unordered gluing configurations. Their exact minimum is 20.

For $(6,7)$, the lower bounds $L(6) = 8$ and $L(7) = 11$ show that only pieces attaining these two minima can possibly give a value below 22. There are 10 and 16 such occurrences, respectively, producing 160 gluings. Their minimum is 23. □

The last finite statement supplies the induction base at order ten when a dominated deletion remains 2-connected.

Proposition 5.3 (Order-ten base certificate)**.** *Let G be a 2-connected graph of order ten, distinct from $K_{10}, K_{10}^2, K_{10}^8$. Then $\eta(G) \geq 20$.*

Proof. Suppose instead that $\eta(G) \leq 19$. By Lemma 2.4, G has a dominated pair u, v. If some dominated deletion is complete, or is one of the two exceptional 2-connected graphs, then Corollary 3.2 and Propositions 3.5 and 3.6 already give a contradiction. If no dominated deletion is 2-connected, then every one is non-2-connected and Lemmas 2.3 and 4.3 and Proposition 5.2 settle the graph. Hence we may choose a dominated pair such that $H = G - u$ is 2-connected, non-complete and non-special. By Lemma 2.2,

$$\eta(H) \leq 17.$$

Since $\eta(H) \leq 17 < 18$, the contrapositive of Lemma 2.4 gives a dominated pair in the nine-vertex graph H. If all dominated deletions of H were complete or non-2-connected, then Lemma 2.3 and the block estimate would force the only possible piece sizes to be $(5,6)$ and hence

$$\eta(H) \geq L(5) + L(6) + 2 = 20,$$

a contradiction. Thus H has a dominated deletion $F = H - a$ that is 2-connected or complete. If F is complete, then $H \cong K_9^t$; by Proposition 3.1, every non-special such graph has $\eta(H) \geq 20$, again a contradiction. Hence F is 2-connected and non-complete, and Lemma 2.2 gives

$$\eta(F) \leq 15.$$

The finite universe can now be generated without an isomorphism search. Start with all 853 connected seven-vertex Atlas representatives and add one non-isolated vertex in every possible way; retain the 2-connected graphs F with $\eta(F) \leq 15$. This produces 97 labelled occurrences. From each, generate every dominated extension from Lemma 2.6; retain the non-special graphs H with $\eta(H) \leq 17$. This produces 681 occurrences. Finally generate every dominated extension of each retained H, excluding only the three special order-ten graphs. At this last stage no 2-connectivity filter and no isomorphism deduplication are used, so the 300311 generated graphs form a superset of all remaining candidates. Their minimum value is exactly 20, a contradiction. □

Remark 5.4 (What the certificate checks). The verifier performs all-pairs breadth-first search, sums the resulting integer distances for W, and for every edge uv counts directly the vertices on the two strict distance sides to obtain the factor $n_{uv}(u)n_{uv}(v)$. A graph is tested for 2-connectivity by deleting each vertex in turn and checking connectivity. The three exceptional families are recognized by their degree sequences, which uniquely determine them:

$$\begin{aligned} K_n &: (n-1)^n, \\ K_n^2 &: 2, (n-2)^{n-3}, (n-1)^2, \\ K_n^{n-2} &: (n-2)^2, (n-1)^{n-2}. \end{aligned}$$

For K_n^2, the two vertices of degree $n-1$ are universal, so the degree-2 vertex is forced to be adjacent exactly to them. For K_n^{n-2}, the $n-2$ vertices of degree $n-1$ are universal, forcing

the remaining two vertices to be non-adjacent. Thus these degree tests introduce no isomorphism ambiguity. All arithmetic is integral. The critical order-ten run intentionally keeps repeated labelled copies, so correctness does not depend on a graph-isomorphism algorithm or a canonical-labelling hash.

6. Proof of the conjecture

Proof of Theorem 1.1. The case $n = 10$ is Proposition 5.3. Suppose $n \geq 11$ and choose a counterexample G of minimum order. The graph is non-complete and is not C_5. By Lemma 2.4, there are vertices u, v with $N[u] \subseteq N[v]$.

Fix any such pair and put $H = G - u$. If H is complete, then Corollary 3.2 gives a contradiction. If H is 2-connected and non-complete, there are three possibilities. If

$$H \ncong K^2_{n-1}, K^{n-3}_{n-1},$$

then H has order $n - 1 \geq 10$ and is not one of the three exceptional graphs for its order. Minimality of G gives

$$\eta(H) \geq 2(n - 1),$$

and Lemma 2.2 gives $\eta(G) \geq 2n$, a contradiction. If $H \cong K^2_{n-1}$ or K^{n-3}_{n-1}, the contradiction follows from Propositions 3.5 and 3.6.

Consequently, for every dominated pair in G, the deletion of the dominated vertex is not 2-connected. The hypotheses of Lemma 2.3 are therefore satisfied. Let $C_1, \ldots, C_k$ be the blocks of $G - u$ and form the pieces G_i as in Section 4. Each G_i is a proper 2-connected subgraph of G, and Lemma 4.1 shows that it is not one of the three exceptional graphs for its order.

If $s_i \geq 10$, minimality therefore gives $\eta(G_i) \geq 2s_i$; if $s_i < 10$, Proposition 5.1 gives the lower bound $L(s_i)$. Hence Lemmas 4.2 and 4.3 apply. For $n \geq 12$ they immediately give $\eta(G) \geq 2n$. If $n = 11$, the only unresolved size pattern is $(6, 7)$, which is settled by Proposition 5.2. This final contradiction proves the theorem. □

7. Certificate and reproducibility

The supplementary archive contains three C++17 verifiers and the fixed small-graph snapshots used in Section 5. The command `verify.sh` checks the SHA256 digests of the inputs, compiles the verifiers, and reproduces the counts and minima stated in Propositions 5.1 to 5.3. The fixed files are `connected7_graph6.txt`, `connected7_masks.txt`, and `atlas_bases_4_7.txt`. An optional Python script regenerates them from NetworkX 3.6.1; NetworkX is not needed once the snapshots are present. The C++ programs use only integer arithmetic and the algorithms described in Remark 5.4.

References

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